

Binomial Distributions

Road Map

Binomial
Distributions

Binomial
Requirements

Sampling Distribution
of a Count

Motivation

So far, we have used a combination of specific rules, such as the

- Addition Rule, Multiplication Rule, Bayes Rule, etc,
- to calculate probabilities and expected values.

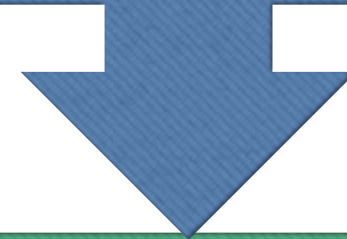
But, it seems a bit out of place, given this class's emphasis on application...

These specific rules can be transformed into probability models,

- which, in turn, generate distributions – not unlike the normal distribution.

Binomial Distributions

Just like the normal distribution describes bell-shaped data, the binomial distribution describes count data.



Specifically, it models the number of successful outcomes,

when the total number of trials is known beforehand.

(The distribution of a count depends on how the data are produced. The binomial setting is a common situation.)

Binomial Distribution



What does a binomial
distribution look like?



It depends

on two parameters

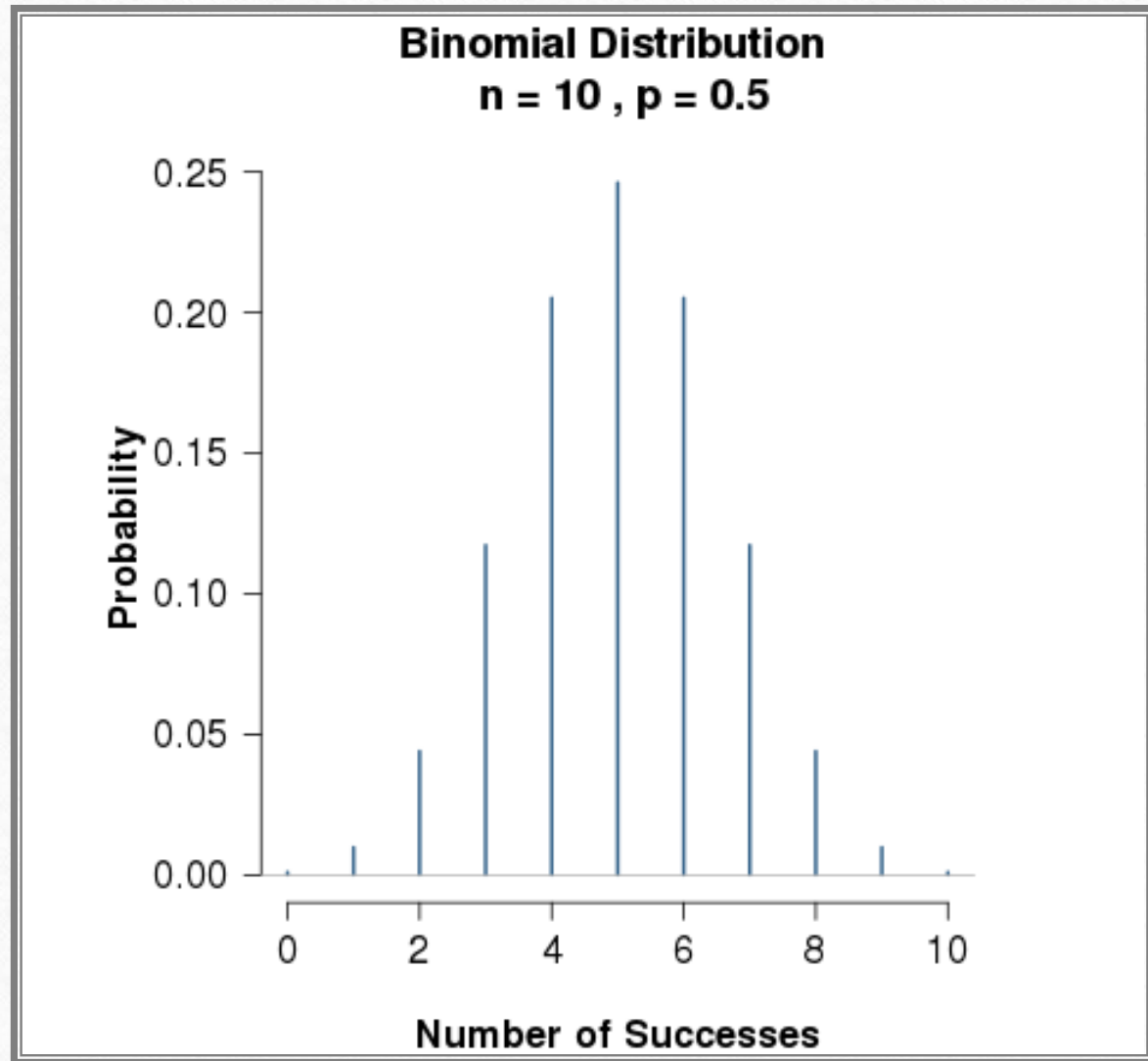
- Probability of Success (P)
- Sample Size/Number of Trials (N)



(Recall that the normal
distribution depends on μ and
 σ)

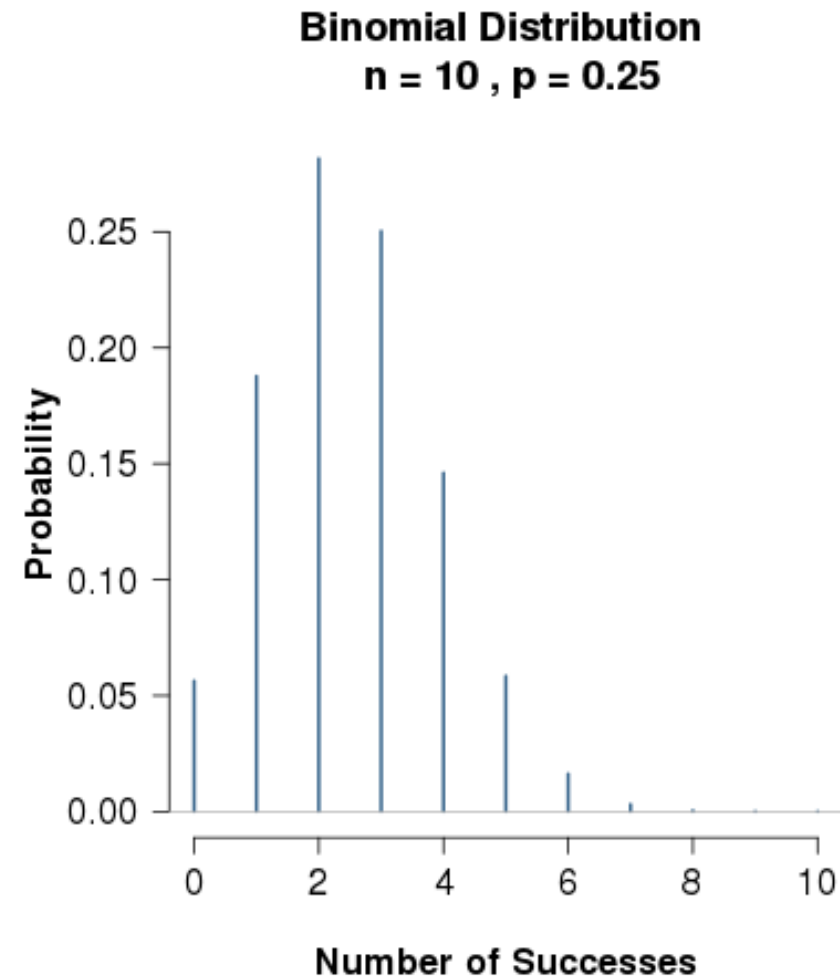
Ten Trials, Probability of Success is 50%

Successes	Probability
0	0.0009766
1	0.0097656
2	0.0439453
3	0.1171875
4	0.2050781
5	0.2460938
6	0.2050781
7	0.1171875
8	0.0439453
9	0.0097656
10	0.0009766



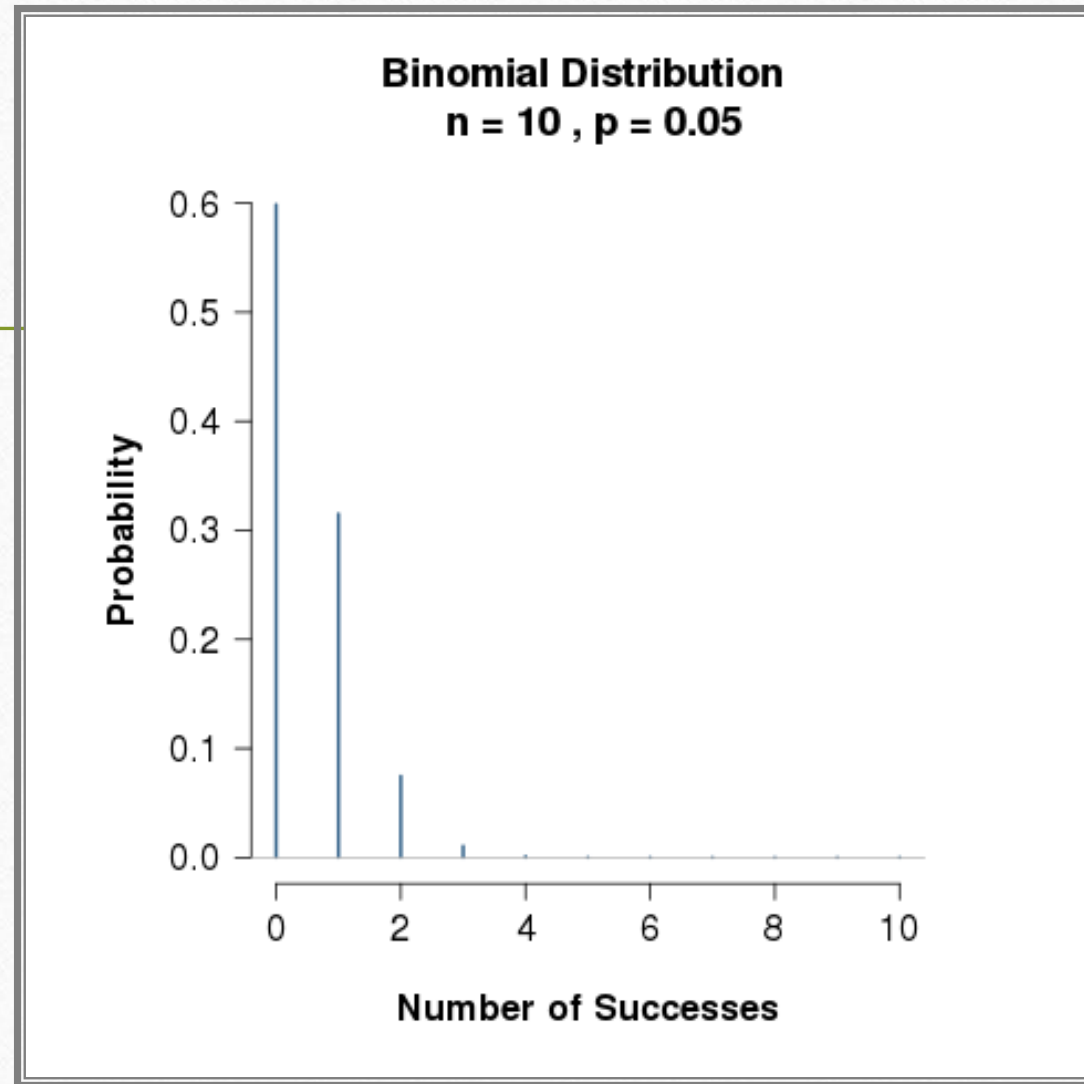
Ten Trials, Probability of Success is 25%

Successes	Probability
0	0.0563135
1	0.1877117
2	0.2815676
3	0.2502823
4	0.1459980
5	0.0583992
6	0.0162220
7	0.0030899
8	0.0003862
9	0.0000286
10	0.0000010



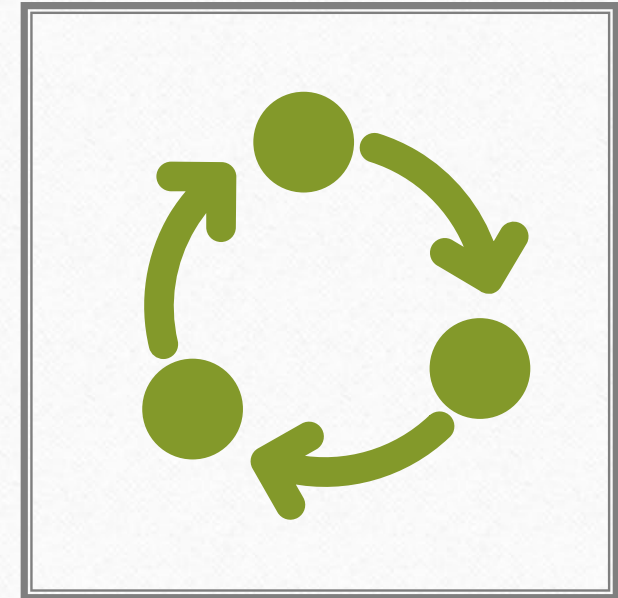
Ten Trials, Probability of Success is 5%

Successes	Probability
0	0.5987369
1	0.3151247
2	0.0746348
3	0.0104751
4	0.0009648
5	0.0000609
6	0.0000027
7	8.038e-8
8	1.586e-9
9	1.855e-11
10	9.766e-14



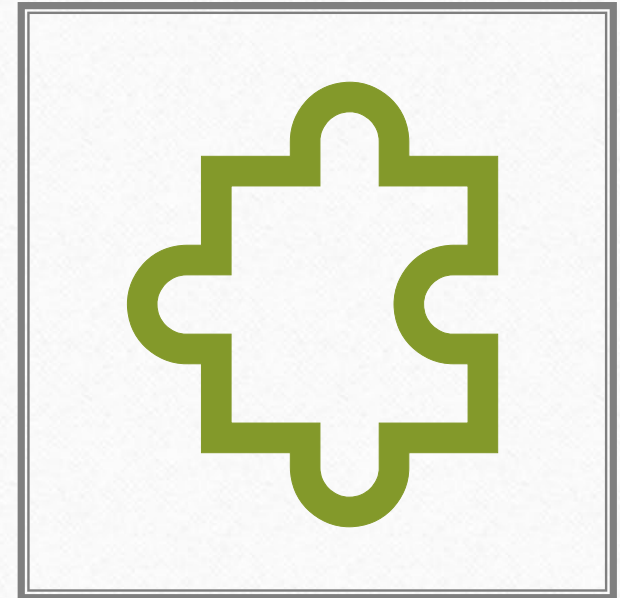
Binomial Distribution

- Note: This is a discrete probability model
 - The distribution isn't smooth because every outcome can fall into a bin
 - i.e, There are a finite number of values to measure.
 - In contrast, a continuous probability model has a smooth range of values.
 - Even when N is large it is still discrete.
 - See <https://shiny.rit.albany.edu/stat/binomial/>



Binomial Requirements

1. There are a fixed number, n , of observations.
2. The n observations are all independent. That is, knowing the result of one observation does not change the probabilities we assign to other observations.
3. Each observation falls into one of just two categories, which, for convenience, we call “success” and “failure.”
4. The probability of a success, call it p , is the same for each observation.





Wrapping Up...

This Time...
N chose K

Formula for Binomial Probabilities

- $P(k) = \frac{N!}{k!(N-k)!} p^k (1-p)^{N-k}$
- where, $P(k)$ is the probability of k successes out of N trials, N is the number of trials and p is the probability of success
- Sometimes this equation is written as
- $P(X=k | pN) = \binom{n}{k} p^k (1-p)^{N-k}$
 - Reads as n chose k

Binomial Formula Breakdown

- The binomial coefficient counts the number of different ways in which k successes can be arranged among n trials.
- The binomial probability $P(X = k)$ is this count multiplied by the probability of any one specific arrangement of the k successes.

BINOMIAL PROBABILITY

- If X has the binomial distribution with n trials and probability p of success on each trial, the possible values of X are $0, 1, 2, \dots, n$. If k is any one of these values:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Number of
arrangements of k
successes

Probability of k
successes

Probability of n
– k failures

N chose K

- Chose K numbers from N options $\binom{n}{k}$
- $\frac{N!}{k!(N-k)!}$
- Note: $0! = 1$
- $K! = K(K-1)(K-2)(K-3) \dots 3 \dots 2 \dots 1$
- $6!/4! = 6 \times 5$

Draw a card, guess its color

- $P=1/2$; $1-p=1/2$
- Let's do $n = 3$
- $\binom{n}{k} p^k (1-p)^{N-k}$

Draw a card, guess its color (0 correct)

- $\frac{N!}{k!(N-k)!} p^k (1-p)^{N-k}$
- $P(x=0 | 1/2, 3) = \binom{3}{0} .5^0 (1-.5)^{3-0}$
- $= \frac{3!}{0!(3-0)!} \cdot .5^0 (1-.5)^{3-0}$
- $= \frac{3!}{1*3!} \cdot .5^0 (.5)^3$
- $= 1 * 1 \quad (.5)^3 = 1/8$

Draw a card, guess its color (1 correct)

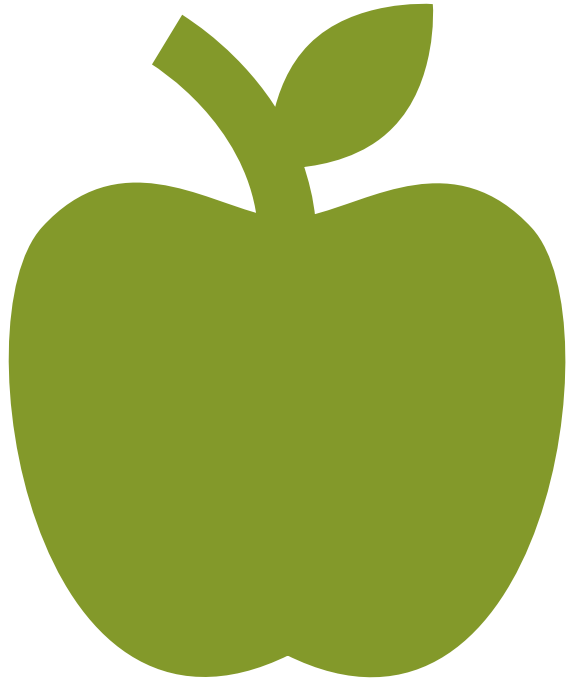
- $\frac{N!}{k!(N-k)!} p^k (1-p)^{N-k}$
- $P(x=1 \mid 1/2, 3) = \binom{3}{1} .5^1 (1-.5)^{3-1}$
- $= \frac{3!}{1!(3-1)!} \cdot .5^1 (1-.5)^2$
- $= \frac{3!}{1*2!} \cdot .5^1 (.5)^2$
- $= 3 * .5^1 (.5)^2 = 3/8$

Draw a card, guess its color (2 correct)

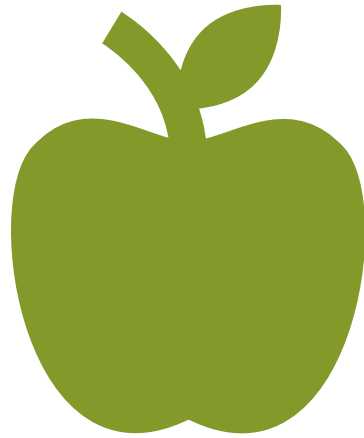
- $\frac{N!}{k!(N-k)!} p^k (1-p)^{N-k}$
- $P(x=2 | 1/2, 3) = \binom{3}{2} .5^2 (1-.5)^{3-2}$
- $= \frac{3!}{2!(3-2)!} .5^2 (1-.5)^1$
- $= \frac{3!}{2! * 1!} .5^2 (.5)^1$
- $= 3 * .5^2 (.5)^1 = 3/8$

Draw a card, guess its color (3 correct)

- $\frac{N!}{k!(N-k)!} p^k (1-p)^{N-k}$
- $P(x=3 | 1/2, 3) = \binom{3}{3} .5^3 (1-.5)^{3-3}$
- $= \frac{3!}{3!(3-3)!} \cdot .5^3 (1-.5)^0$
- $= \frac{3!}{3!} \cdot .5^3 (.5)^0$
- $= 1 * .5^3 = 1/8$



Wrapping Up...



This Time... Unmarketable tomatoes

Binomial distributions in statistical sampling

- The binomial distributions are important in statistics when we want to make inferences about the proportion p of successes in a population.
- Suppose 11% of one producer's tomatoes are unmarketable. At the warehouse, an official inspects a simple random sample of 10 tomatoes from a shipment of 10,000. Let X = number of unmarketable tomatoes. What is $P(X=0)$?

Example

- Suppose 11% of one producer's tomatoes are unmarketable. At the warehouse, an official inspects a simple random sample of 10 tomatoes from a shipment of 10,000. Let X = number of unmarketable tomatoes.
- What is $P(X=0)$?
- Without the binomial, we could calculate the probability like this:

$$\frac{8900}{10000} \cdot \frac{8899}{9999} \cdot \frac{8898}{9998} \cdots \frac{8891}{9991} = 0.3116$$

Example

- Suppose 11% of one producer's tomatoes are unmarketable. At the warehouse, an official inspects a simple random sample of 10 tomatoes from a shipment of 10,000. Let X = number of unmarketable tomatoes.
- What is $P(X=0)$?
- With the binomial, we could calculate the probability like this:
- $$P(X = 0) \approx \binom{10}{0} (0.11)^0 0.89^{10} = 0.3118$$

Sampling Distribution of a Count

- Choose an SRS of size n from a population with proportion p of successes.
- When the sample size is less than 5% of the population size, the count X of successes in the sample has approximately the binomial distribution with parameters n and p .

Binomial mean and standard deviation

- If a count X has the binomial distribution based on n observations with probability p of success, what is its mean, μ ?
- If a count X has the binomial distribution with number of trials n and probability of success p , the mean and standard deviation of X are:

$$\mu = np$$

$$\sigma = \sqrt{np(1 - p)}$$

Recall our tomatoes

- 11% of one producer's tomatoes are unmarketable. At the warehouse, an official inspects a simple random sample of 10 tomatoes from a shipment of 10,000.
- What would the mean and standard deviation be for the binomial distribution of these 10 tomatoes.

$$\mu = np = 10 * .11 = 1.1$$

$$\sigma = \sqrt{np(1 - p)} = \sqrt{10 * .11(1 - .11)} = \sqrt{1.1(1 - .11)} = 0.989$$



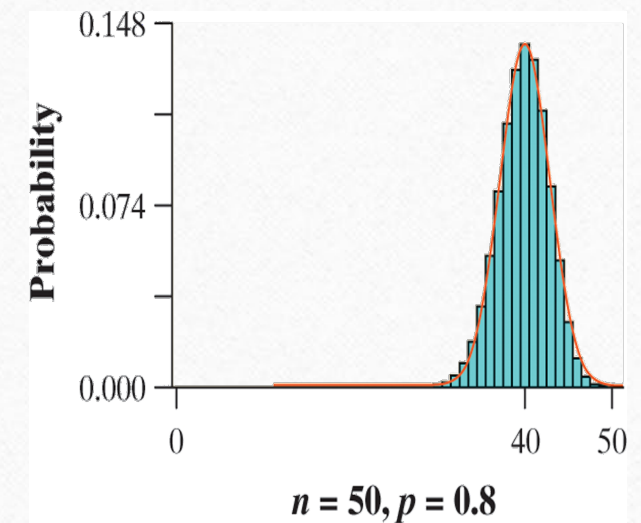
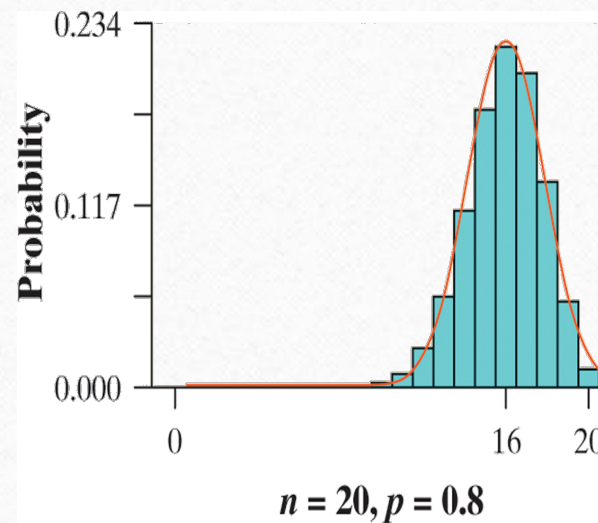
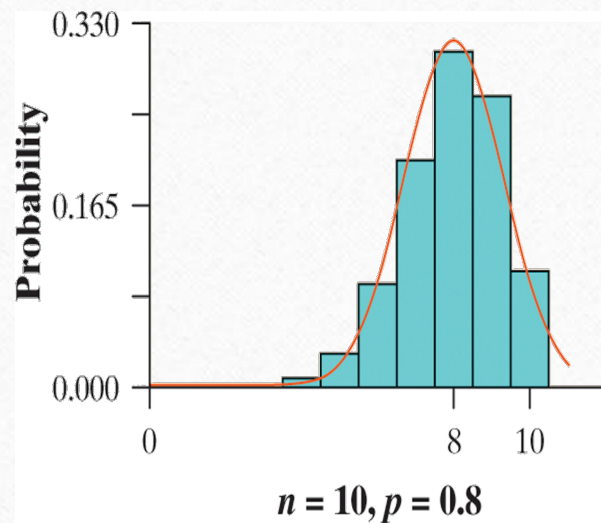
Wrapping Up...



This Time...
Normal Approximation

Normal Approximation

- As n gets larger, something interesting happens to the shape of a binomial distribution.



Normal Approximation for Binomial Distributions

Suppose that a count X has the binomial distribution with n observations and success probability p .

When n is large, the distribution of X is approximately Normal,

- $N(np, \sqrt{np(1-p)})$.

As a rule of thumb, we will use the Normal approximation when n is so large that $np \geq 10$ and $n(1-p) \geq 10$.



Wrapping Up...

Functions in R

`dbinom()` gives the density,

`pbinom(q, size, prob...)` gives the cumulative density function,

- computes the probability that a binomially distributed random number will be at that value or less than that number

`qbinom()` gives the quantile function, and

- (is the inverse of `pbinom`, give it a probability,
- it produces the number whose cumulative distribution matches the probability

`rbinom()` generates random deviates.

Examples using R

- **Example**
- **Question:** Suppose widgets produced at Acme Widget Works have probability 0.005 of being defective. Suppose widgets are shipped in cartons containing 25 widgets. What is the probability that a randomly chosen carton contains exactly one defective widget?

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- **Question Rephrased:** What is $P(X = 1)$ when X has the $\text{Bin}(25, 0.005)$ distribution?
 - `dbinom(1, 25, 0.005)` [1] 0.1108317

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- **Question:** Suppose widgets produced at Acme Widget Works have probability 0.005 of being defective. Suppose widgets are shipped in cartons containing 25 widgets. What is the probability that a randomly chosen carton contains no more than one defective widget?

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- **Question Rephrased:** What is $P(X \leq 1)$ when X has the $\text{Bin}(25, 0.005)$ distribution?
 - `pbinom(1, 25, 0.005)` [1] 0.9930519

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- **Question:** What are the 10th, 20th, and so forth quantiles of the $\text{Bin}(10, 1/3)$ distribution?
 - `> qbinom(0.1, 10, 1/3) [1] 1`
 - `> qbinom(0.2, 10, 1/3) [1] 2`
 - `> # and so forth, or all at once with`
 - `> qbinom(seq(0.1, 0.9, 0.1), 10, 1/3)`
 - `[1] 1 2 3 3 3 4 4 5 5`

Wrapping
Up...
